POWER PULSE

Per Unit System Important Formulae

Notes by Balaji Sir

1 Definitions

1.
$$\text{Quantity (pu)} = \frac{\text{actual value}}{\text{base value}}$$

2.
$$Z_{base} = \frac{kV_{base}^2}{MVA_{base}}\ (\Omega), \qquad I_{base} = \frac{MVA_{base}}{\sqrt{3}\,kV_{base}}\ (\text{kA})$$

kV: line-to-line, MVA: 3-phase

3.
$$Z_{pu} = \frac{Z(\Omega)}{Z_{base}} = \frac{Z(\Omega)\,MVA_{base}}{kV_{base}^2}$$

2 Change of base

4.
$$Z_{pu,new} = Z_{pu,old}\left(\frac{kV_{old}}{kV_{new}}\right)^2\left(\frac{MVA_{new}}{MVA_{old}}\right)$$

3 Multi-voltage systems

5.
$$kV_{base,2} = kV_{base,1} \times \frac{\text{rated kV of side 2}}{\text{rated kV of side 1}}$$

base voltages follow the transformer ratio

6. Z_{pu} of a transformer is the same on either side

7.
$$\text{Transformer: } Z_{pu} = \frac{\%Z}{100} = \frac{V_{sc}}{V_{rated}}$$

4 Fault quantities

8.
$$I_{f,pu} = \frac{1}{Z_{pu}}, \qquad I_f = I_{base}\,I_{f,pu}$$

9.
$$S_{sc} = \frac{MVA_{base}}{Z_{pu}}$$

10. Advantages: no voltage-level conversion, transformer ratio disappears, values lie near 1.0

11.
$$V_{pu} \approx 1.0, \qquad S_{pu} = \frac{S}{S_{base}}$$